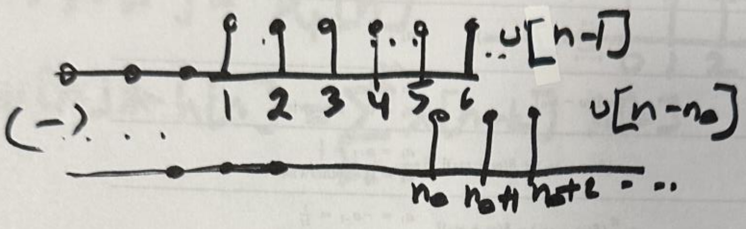
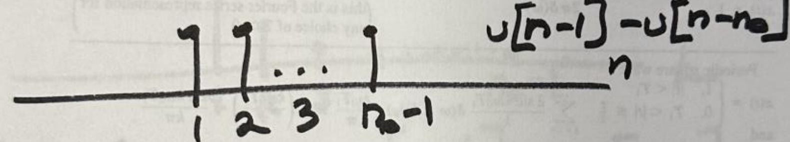
HW 4 SOLUTIONS

1

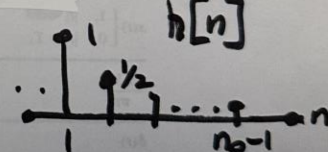
1) $h[n] = \left(\frac{1}{2}\right)^{n-1}\left(u[n-1] - u[n-n_0]\right)$ $\quad n_0 > 4$





a) For stability $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$

$$h[n] = \begin{cases} \left(\frac{1}{2}\right)^{n-1} & 1 \leq n \leq n_0 - 1 \\ 0 & \text{otherwise} \end{cases}$$



$$B_h = \sum_{n=-\infty}^{\infty} |h[n]| = \sum_{n=1}^{n_0-1} \left|\left(\frac{1}{2}\right)^{n-1}\right| = \sum_{m=0}^{n_0-2} \left(\frac{1}{2}\right)^m = \frac{1 - \left(\frac{1}{2}\right)^{n_0-1}}{1 - \frac{1}{2}}$$

Let $m = n-1$

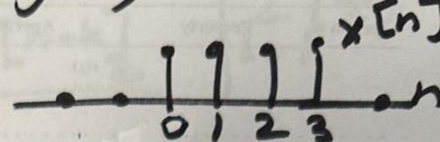
$$= 2 - 2 \cdot \left(\frac{1}{2}\right)^{n_0}\left(\frac{1}{2}\right)^{-1} = 2 - 4\left(\frac{1}{2}\right)^{n_0} \quad \text{(monotone increasing in } n_0\text{)}$$

Let $n_0 = 5$ $B_h = \frac{15}{8}$, let $n_0 \to \infty$ $B_h = 2$

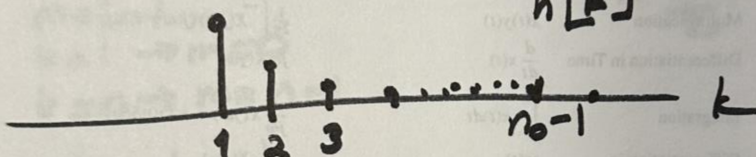
system is stable

1 b) $h[n] = 0$ for $n < 0$ so it is causal.
(but it is not memoryless)

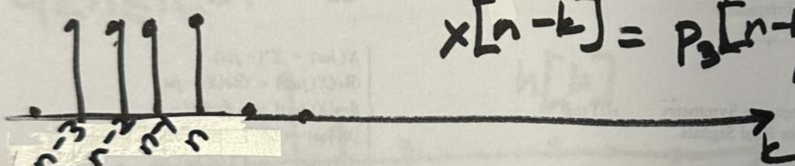
1 c) i) $x[n] = p_4[n]$



$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[n-k]\, h[k]$$



$x[n-k] = p_3[n-k]$

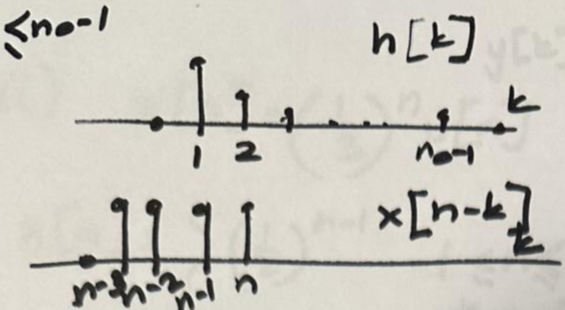


Note:
$n-k=0 \Rightarrow k=n$

$$y[n] = \begin{cases} 0 & n < 1 \\ \sum_{k=1}^{n} \left(\frac{1}{2}\right)^{k-1} & 1 \le n \le 4 \\ \sum_{n-3}^{n} \left(\frac{1}{2}\right)^{k-1} & 4 < n \le n_0 - 1 \\ \sum_{k=n-3}^{n_0-1} \left(\frac{1}{2}\right)^{k-1} & n_0 \le n \le n_0 + 2 \\ 0 & n > n_0 + 2 \end{cases}$$

Let $1 \le n \le n_0 - 1$

3



$\Big\} \sum_{k=1}^{n} \left(\frac{1}{2}\right)^{k-1}$

$$\sum_{k=1}^{n} \left(\frac{1}{2}\right)^{k-1} = \sum_{m=0}^{n-1} \left(\frac{1}{2}\right)^{m} = \frac{1-\left(\frac{1}{2}\right)^{n}}{1-\frac{1}{2}} = 2-\left(\frac{1}{2}\right)^{n-1}$$

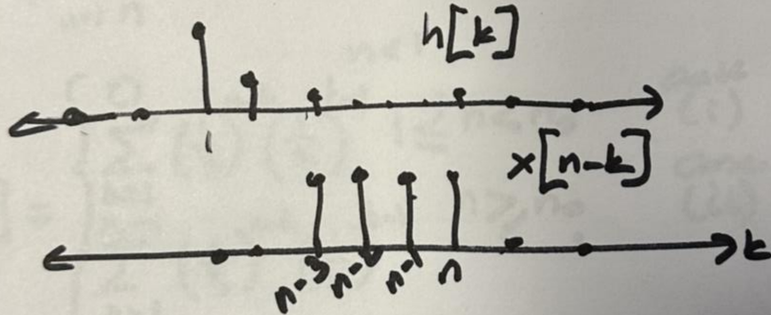
$1 \le n \le 4$

$m = k-1$

$k=1 \rightarrow m=0$

$k=n \rightarrow m=n-1$

Let $4 < n \le n_0 - 1$



$$\sum_{n-3}^{n} \left(\frac{1}{2}\right)^{k-1} = \sum_{m=0}^{3} \left(\frac{1}{2}\right)^{m+n-4} = \left(\frac{1}{2}\right)^{n-4} \left(\frac{1-\left(\frac{1}{2}\right)^{4}}{1-\frac{1}{2}}\right)$$

$m = k-(n-3) = k-n+3$ so $k-1 = m+n-4$

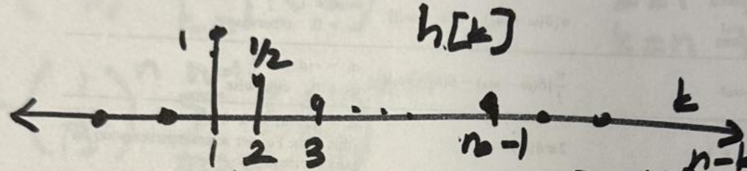
$$= \left(\frac{1}{2}\right)^{n-4} \times \left(2 - 2\left(\frac{1}{2}\right)^{4}\right) = 2\left[\left(\frac{1}{2}\right)^{n-4} - \left(\frac{1}{2}\right)^{n}\right]$$

$4 < n \le n_0 - 1$

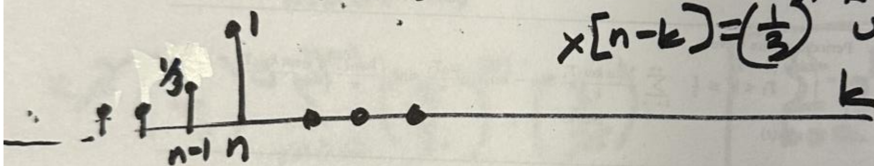
1c (ii) $x[n]=\left(\frac{1}{3}\right)^n u[n]$ $\qquad y[k]=\sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$

4

$$h[n]=\begin{cases}\left(\frac{1}{2}\right)^{n-1} & 1\le n\le n_0-1\\ 0 & \text{otherwise.}\end{cases}$$

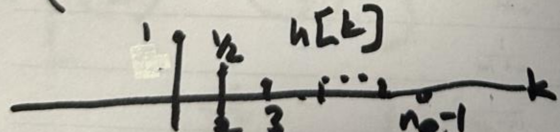


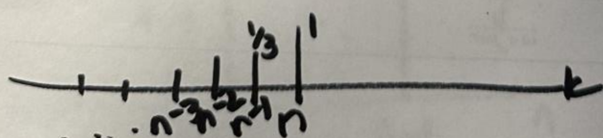
$$x[n-k]=\left(\frac{1}{3}\right)^{n-k}u[n-k]$$



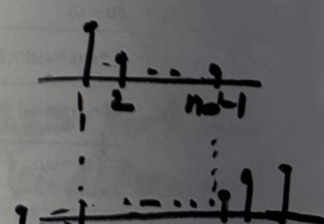
$$y[n]=\begin{cases}0 & n<1\\ \sum_{k=1}^{n}\left(\frac{1}{3}\right)^{n-k}\left(\frac{1}{2}\right)^{k-1} & 1\le n< n_0 \quad \text{case (i)}\\ \sum_{k=1}^{n_0-1}\left(\frac{1}{3}\right)^{n-k}\left(\frac{1}{2}\right)^{k-1} & n\ge n_0. \quad \text{case (ii)}\end{cases}$$

(i)





(ii)



$$\sum_{k=1}^{n}\left(\frac{1}{3}\right)^{n-k}\left(\frac{1}{2}\right)^{k-1} \qquad 1 \le n < n_0$$

$$= \sum_{k=1}^{n}\left(\frac{1}{3}\right)^{n}\left(\frac{1}{3}\right)^{-k}\left(\frac{1}{2}\right)^{k}\left(\frac{1}{2}\right)^{-1}$$

$$= 2\left(\frac{1}{3}\right)^{n}\sum_{k=1}^{n}\left(\frac{3}{2}\right)^{k}$$

Let $m = k-1$
$k=1 \rightarrow m=0$
$k=n \rightarrow m=n-1$

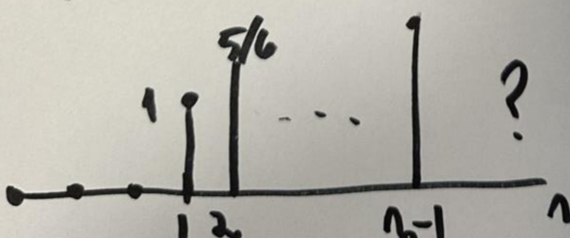
$$= 2\left(\frac{1}{3}\right)^{n}\sum_{m=0}^{n-1}\left(\frac{3}{2}\right)^{m+1}$$

$$= \cancel{2}\left(\frac{1}{3}\right)^{n}\sum_{m=0}^{n-1}\left(\frac{3}{2}\right)^{m}\cdot\left(\frac{3}{\cancel{2}}\right)$$

$$\boxed{\sum_{k=0}^{n} a^k = \frac{1-a^{n+1}}{1-a} \quad \text{for } |a|<1}$$

$$= 3\cdot\left(\frac{1}{3}\right)^{n}\frac{1-\left(\frac{3}{2}\right)^{n}}{1-\frac{3}{2}}$$

$$= 6\left[\left(\frac{1}{2}\right)^{n} - \left(\frac{1}{3}\right)^{n}\right] \qquad 1 \le n < n_0$$



This part will be determined for $n \geq n_0$

Cont (ii)

$$\sum_{k=1}^{n_0-1} \left(\frac{1}{3}\right)^{n-k} \cdot \left(\frac{1}{2}\right)^{k-1} = 2\left(\frac{1}{3}\right)^{n} \sum_{k=1}^{n_0-1} \left(\frac{3}{2}\right)^{k}$$

$$= 2\left(\frac{1}{3}\right)^{n} \sum_{m=0}^{n_0-2} \left(\frac{3}{2}\right)^{k}$$

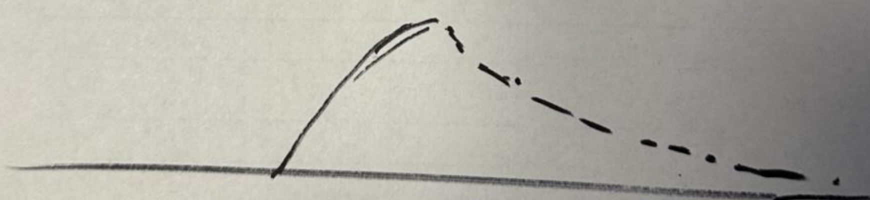
$m=k-1$

$k=n_0-1 \quad m=n_0-2$

$$= 2\left(\frac{1}{3}\right)^{n} \frac{1-\left(\frac{3}{2}\right)^{n_0-1}}{1-3/2} = 4\left(\frac{1}{3}\right)^{n}\left[\left(\frac{3}{2}\right)^{n_0-1} - 1\right]$$

$$= 4\,\frac{3^{n_0-n}}{2^{n_0}} \cdot \frac{2}{3} - \left(\frac{1}{3}\right)^{n}$$

$n \geq n_0$

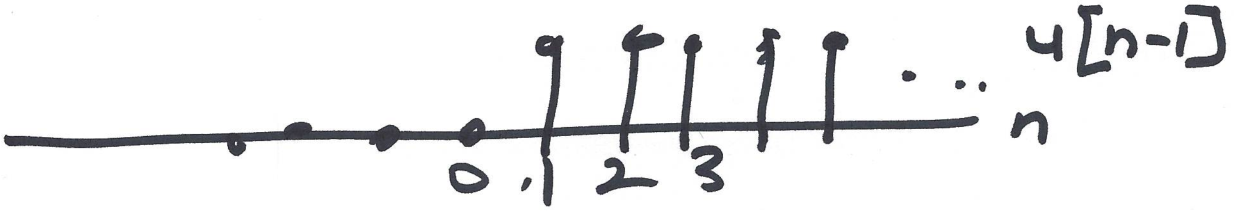


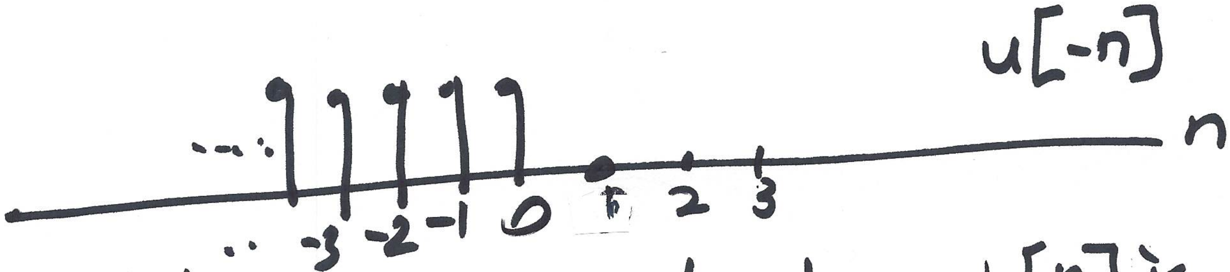
SOLUTIONS: HW 4

2a) $h[n] = \underbrace{\left(\frac{1}{2}\right)^{n-1} u[n-1]}_{h_1[n]} \underbrace{- (-2)^n u[-n]}_{h_2[n]}$

(1)

$h[n] = h_1[n] + h_2[n]$





a) stability: We have to show $h[n]$ is absolutely summable, i.e. $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$

i) Take $h_1[n] = \left(\frac{1}{2}\right)^{n-1} u[n-1]$

$$\sum_{n=-\infty}^{\infty} |h_1[n]| = \sum_{n=1}^{\infty} \left|\left(\frac{1}{2}\right)^{n-1}\right| = \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k = \frac{1}{1-\frac{1}{2}}$$

let $k = n-1$, $= 2$

$$\sum_{n=-\infty}^{\infty} |h_2[n]| = \sum_{n=-\infty}^{0} |-(-2)^n| = \sum_{k=\infty}^{0} |(-2)^{-k}| = \sum_{k=0}^{\infty} \left|\left(\frac{-1}{2}\right)^k\right|$$

$k = -n$

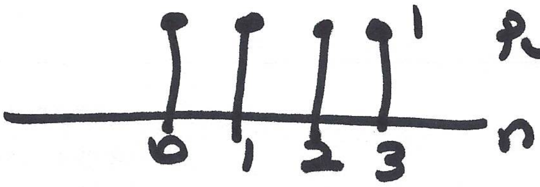
$$= \sum_{k=0}^{\infty} \left|\left(\frac{1}{2}\right)^k\right| = 2$$

$2 + 2 = 4 < \infty$

stable

b) Stable? No because $h_2[n] = -(-2)^n u[-n]$ is nonzero for $n<0$.

c) $x[n] = p_4[n]$



$p_4[n]$

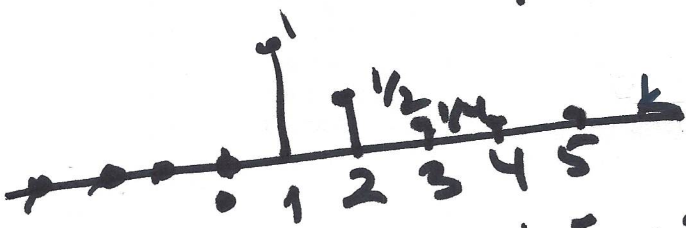
$$y[n] = x[n] * \left[h_1[n] + h_2[n]\right]$$
$$= x[n] * h_1[n] + x[n] * h_2[n]$$
$$= y_1[n] + y_2[n]$$

<u>Concentrate on $y_1[n]$</u> $= \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$

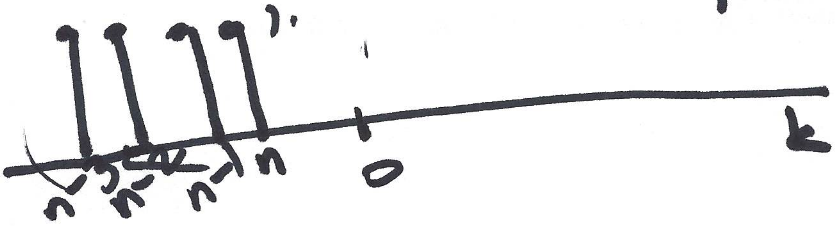
$$= \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$h_1[n] = \left(\tfrac{1}{2}\right)^{n-1} u[n-1]$$

$h_1[k]$



$x[n-k]$
$= p_4[n-k]$



$$y_1[n] = \begin{cases} 0 & n \le 0 \\ \sum_{k=1}^{n} \left(\tfrac{1}{2}\right)^{k-1} & 1 \le n \le 4 \\ \sum_{k=n-3}^{n} \left(\tfrac{1}{2}\right)^{k-1} & n > 4 \end{cases}$$

$$\sum_{k=1}^{n}\left(\frac{1}{2}\right)^{k-1} = \sum_{\substack{m=0\\ m=k-1}}^{n-1}\left(\frac{1}{2}\right)^{m} = \frac{1-\left(\frac{1}{2}\right)^{n}}{1-\frac{1}{2}} = 2\left[1-\left(\frac{1}{2}\right)^{n}\right]$$

$1 \le n \le 4$

$$\sum_{k=n-3}^{n}\left(\frac{1}{2}\right)^{k-1} = \sum_{m=0}^{3}\left(\frac{1}{2}\right)^{m+n-4} = \left(\frac{1}{2}\right)^{n}2^{4}\,\frac{1-\left(\frac{1}{2}\right)^{4}}{1-\frac{1}{2}}$$

$m = k-(n-3)$

$k=n-3 \rightarrow m=0$

$k=n \rightarrow m = n-(n-3) = 3$

$m+(n-3)-1 = k-1$

$$= \left(\frac{1}{2}\right)^{n}\frac{15}{1/2} = 30\left(\frac{1}{2}\right)^{n}$$

$n \geq 4$

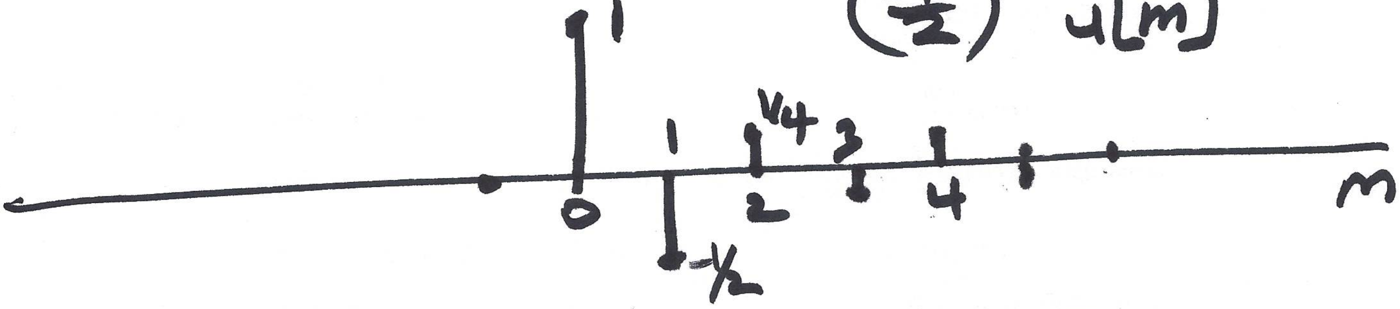
<u>Consider $y_2[n]$</u>

$$y_2[n] = x[n] * h_2[n]$$
$$= p_4[n] * \left[(-2)^{n}u[n]\right]$$

(4)

$$y_2[n] = x[n] * h_2[n]$$
$$= p_4[n] * \left(-(-2)^n u[-n]\right)$$
$$= \sum_{k=-\infty}^{\infty} p_4[n-k] \left(-(-2)^k u[-k]\right) \qquad m=-k$$
$$= \sum_{m=\infty}^{-\infty} p_4[n+m] \left[-(-2)^{-m} u[m]\right)$$
$$= -\sum_{m=0}^{\infty} p_4[n+m] \left(-\tfrac{1}{2}\right)^m$$

$\left(-\frac{1}{2}\right)^m u[m]$



$p_4[n+m]$

$$y_2[n] = \begin{cases} 0 & 3-n<0 \text{ or } n>3 \\ -\sum_{m=0}^{3-n} \left(-\frac{1}{2}\right)^m & 3 \geq n \geq 0 \quad (i) \\ -\sum_{-n}^{3-n} \left(-\frac{1}{2}\right)^m & n<0 \quad (ii) \end{cases}$$

Case i)

$$-\sum_{m=0}^{3-n}\left(\frac{-1}{2}\right)^m = -\frac{1-\left(\frac{-1}{2}\right)^{4-n}}{1-\frac{-1}{2}} = -\frac{2}{3}\left[1-\frac{1}{16}(-2)^n\right]$$

5

$$3 \geq n \geq 0$$

Case ii)

$$-\sum_{m=-n}^{3-n}\left(\frac{-1}{2}\right)^m = -\sum_{k=0}^{3}\left(\frac{-1}{2}\right)^{k-n} = -(-2)^n\left[\frac{1-\left(\frac{-1}{2}\right)^4}{1+\frac{1}{2}}\right]$$

$k = m+n$

$m=-n \rightarrow k=0$

$m=3-n \rightarrow k=3$

$$= -(-2)^n\frac{1-1/16}{3/2} = -(-2)^n\frac{\frac{15}{16}}{3/2}$$

$$= -\frac{5}{8}(-2)^n \qquad n<0$$

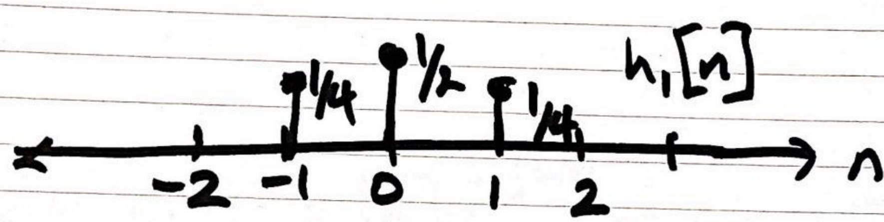
$$y[n]=y_1[n]+y_2[n]$$

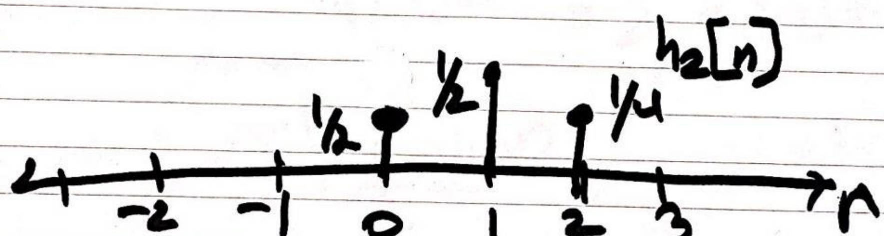
$$y[n]=\begin{cases} -\frac{5}{8}(-2)^n & n<0 \\ 2\left[1-\left(\frac{1}{2}\right)^n\right]-\frac{2}{3}\left[1-\frac{1}{16}(-2)^n\right] & 3\geq n\geq 0 \\ 30\left(\frac{1}{2}\right)^n & n\geq 4 \end{cases}$$

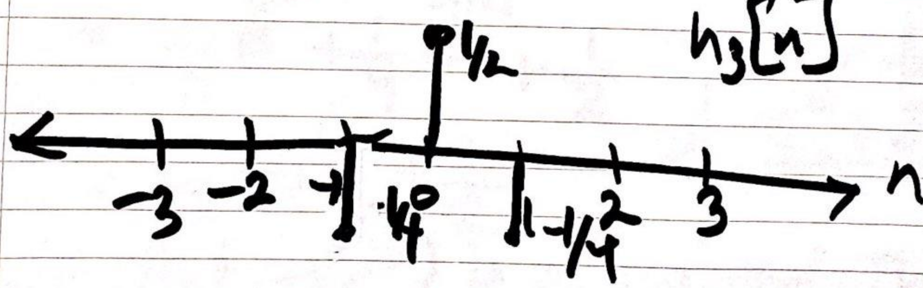
3 a) $h_1[n] = \frac{1}{4}\delta[n-1] + \frac{1}{2}\delta[n] + \frac{1}{4}\delta[n+1]$

$h_2[n] = \frac{1}{4}\delta[n-2] + \frac{1}{2}\delta[n-1] + \frac{1}{4}\delta[n]$

$h_3[n] = -\frac{1}{4}\delta[n-1] + \frac{1}{2}\delta[n] - \frac{1}{4}\delta[n+1]$







②

$$H_1(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h_1[n] e^{-j\omega n} = \sum_{n=-1}^{1} h_1[n] e^{-j\omega n}$$

$$= \frac{1}{4} e^{j\omega 1} + \frac{1}{2} e^{j\omega 0} + \frac{1}{4} e^{-j\omega 1}$$

$$= \frac{1}{4} e^{j\omega} + \frac{1}{4} e^{-j\omega} + \frac{1}{2}$$

euler formula $\frac{e^{j\omega} + e^{-j\omega}}{2} = \cos\omega$

$$= \frac{1}{2} \cos\omega + \frac{1}{2}$$

Note that $H(e^{j\omega})$ is purely real. This is a consequence of the fact that $h_1[n]$ was symmetric.

Even symmetric in time → purely real in freq.

[Odd symmetric in time → purely imaginary in freq.]

$$H_2(e^{j\omega}) = \frac{1}{4}e^{j\omega 2} + \frac{1}{2}e^{j\omega 1} + \frac{1}{4}e^{j\omega 0}$$

Recall $h_2[n] = h_1[n-1]$

$$= e^{j\omega}\underbrace{\left[\frac{1}{4}e^{j\omega} + \frac{1}{2} + \frac{1}{4}\cdot e^{-j\omega}\right]}$$

$$= e^{j\omega} H_1(e^{j\omega})$$

$$= e^{j\omega}\left(\frac{1}{2}\cos\omega + \frac{1}{2}\right)$$

Shift in time by n_0 is multiplication by $e^{-j\omega n_0}$. Here $n_0 = 1$

$$H_3(e^{j\omega}) = \frac{-1}{4}e^{j\omega} + \frac{1}{2}e^{j\omega 0} - \frac{1}{4}e^{-j\omega}$$

$$= \frac{1}{2} - \frac{1}{2}\cos\omega$$

Note that $H_3(e^{j\omega})$ is also purely real. Again this is due to the fact that $h_3[n]$ was symmetric.

symmetric $\longleftrightarrow$ real

c-d)
i) $|H(e^{j\omega})| = \frac{1}{2} + \frac{1}{2}\cos\omega \qquad -\pi \leq \omega < \pi$

$\angle H(e^{j\omega}) = 0$

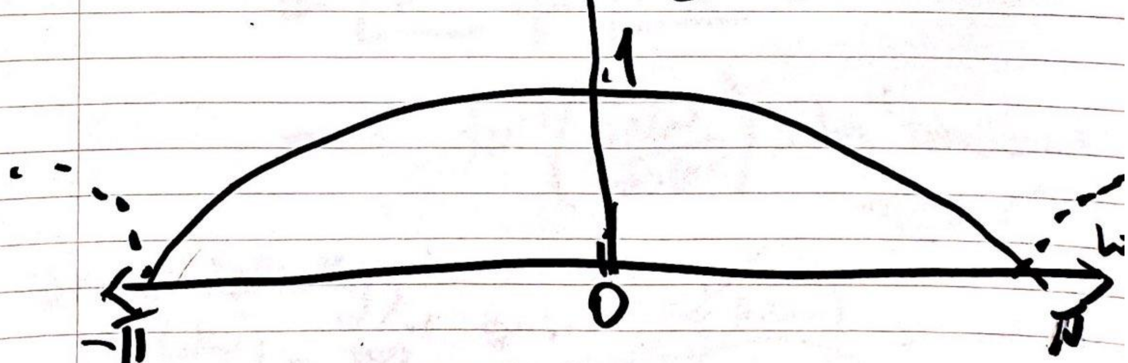
Remember: $\angle H(e^{j\omega}) = \tan^{-1}\frac{\text{Im}(H(e^{j\omega}))}{\text{Re}(H(e^{j\omega}))}$

$= \tan^{-1}(0) = 0 \quad -\pi \leq \omega < \pi$

Note that $H(e^{j\omega})$ is always positive for this example because $\frac{1}{2}+\frac{1}{2}\cos\omega \geq 0$

For a purely real $H(e^{j\omega})$, only phase values $\angle H(e^{j\omega}) = 0$ or $-\pi$ where $-\pi$ corresponds to a negative value of $H(e^{j\omega})$.

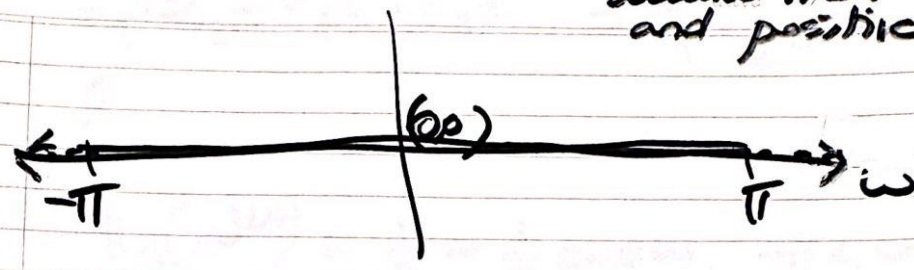
$|H(e^{j\omega})| = \frac{1}{2}+\frac{1}{2}\cos\omega$



This is a low pass filter.

(5)

$\angle H_1(e^{j\omega})$



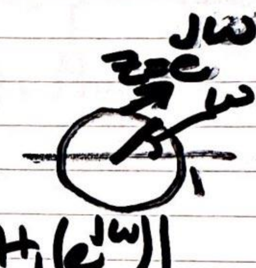
"zero phase" because it's real and positive.

c-d)

ii) $H_2(e^{j\omega}) = e^{j\omega} H_1(e^{j\omega}) = (\cos\omega + j\sin\omega)\left(\frac{1}{2}+\frac{1}{2}\cos\omega\right)$

$|H_2(e^{j\omega})| = |e^{j\omega} H_1(e^{j\omega})|$

$= |e^{j\omega}| \cdot |H_1(e^{j\omega})|$



$= 1 \cdot \left(\frac{1}{2}+\frac{1}{2}\cos\omega\right) = |H_1(e^{j\omega})|$

$\angle H_2(e^{j\omega}) = \underline{\angle e^{j\omega} H_1(e^{j\omega})}$

$= \angle e^{j\omega} + \angle H_1(e^{j\omega})$

$= \tan^{-1}\left(\frac{\sin\omega}{\cos\omega}\right) = \tan^{-1}\tan\omega = \omega$

$-\pi \leq \omega < \pi$

Or equivalently,

$\angle H_2(e^{j\omega}) = \tan^{-1} \frac{\sin\omega\,\cancel{\left(\frac{1}{2}+\frac{1}{2}\cos\omega\right)}}{\cos\omega\,\cancel{\left(\frac{1}{2}+\frac{1}{2}\cos\omega\right)}} = \omega$

$-\pi \leq \omega < \pi$

$= \tan^{-1}(\tan\omega) = \omega$

Reminder: $z_1, z_2 \in \mathbb{C}$

$$z_1 \cdot z_2 = r_1 e^{j\omega_1} \cdot r_2 e^{j\omega_2} = r_1 \cdot r_2 e^{j(\omega_1+\omega_2)}$$

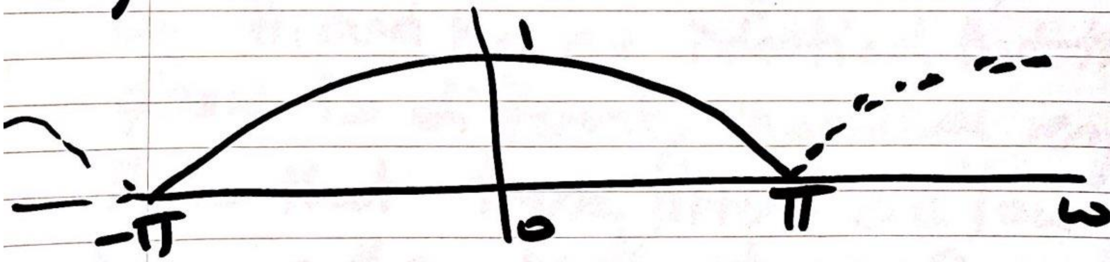
where $r_1 = |z_1|$ and $r_2 = |z_2|$

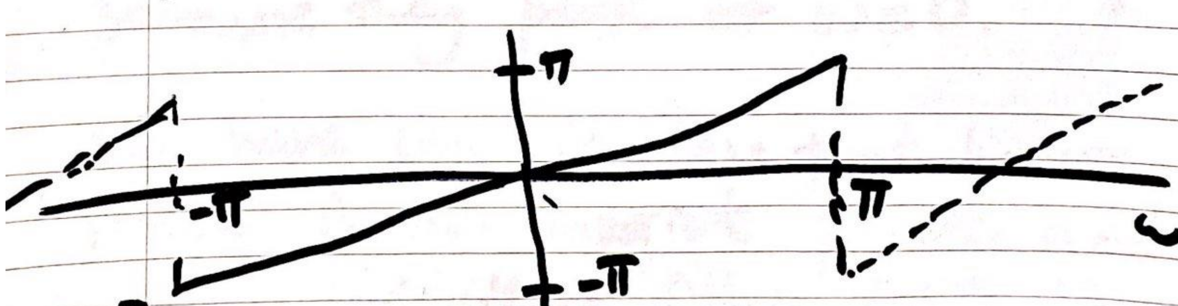
5c-d) ii continued: We showed that

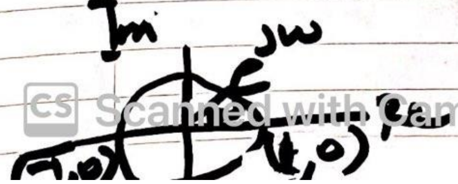
$$|H_2(e^{j\omega})| = |H_1(e^{j\omega})| \qquad -\pi<\omega<\pi$$

$$\angle H_2(e^{j\omega}) = \omega \qquad -\pi<\omega<\pi$$

Moreover, $H(e^{j\omega})$ is periodic with period 2π like any other Discrete time sequence







Reminder:

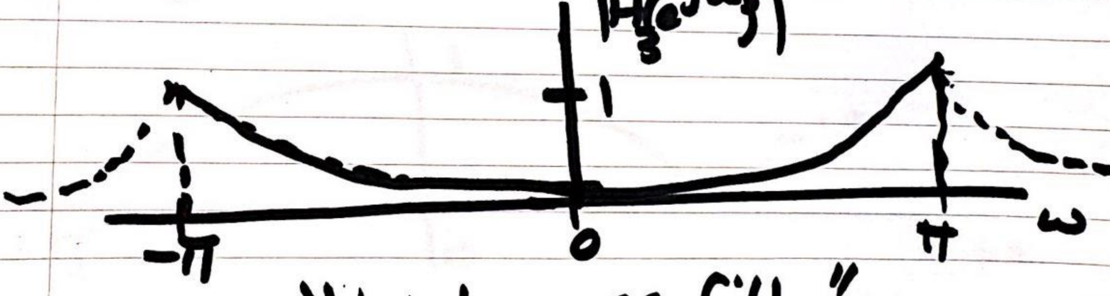
7

$$z_1 \cdot z_2 = r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 \cdot r_2 e^{j(\theta_1+\theta_2)}$$

c-d)
(ii) $H_3(e^{j\omega}) = \frac{1}{2} - \frac{1}{2}\cos\omega \quad -\pi \leq \omega < \pi$

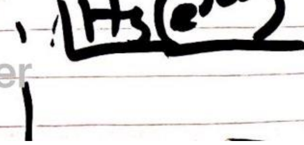
$$|H_3(e^{j\omega})| = \left|\frac{1}{2} - \frac{1}{2}\cos\omega\right|$$

Note that $H_3(e^{j\omega}) \geq 0$ for this particular example so $|H_3(e^{j\omega})| = H(e^{j\omega})$. Normally, we would have to compute the $|\cdot|$ but there is no need here.



"high-pass filter"

$\angle H_3(e^{j\omega}) = 0$ because it is purely real and positive.



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e) All of these "Finite Impulse Filters" have the same coefficients. $h_1[n]$ and $h_2[n]$ differ only by a delay of $n_0=1$ unit. This delay manifests itself as "multiplication by $e^{-j\omega \cdot 1}$" in the frequency domain, i.e. the result is a phase change but the magnitudes are the same.

$$|H_1(e^{j\omega})| = |H_2(e^{j\omega})|$$

Notice that the magnitude spectra of H_1 and H_2 are identical but their phase is different. Magnitude responses show that these filters are "low-pass" because they peak at $\omega=0$.

We have seen in class that linear phase is an important property and we see that these filters satisfy linear phase.

Comparing $h_1[n]$ and $h_3[n]$, the filter coefficients are such that the odd filter elements alternate, ie

$$h_3[n] = [-1]^n h_1[n]$$

This alternation makes the filter $h_3[n]$ a "high pass filter" because the minimum magnitude is at $\omega=$ (origin) and the highest value is at $\omega = \pm\pi$ which is the high frequency. Also note that with this choice of coefficients $H_1(e^{j\omega}) + H_3(e^{j\omega}) =$

On another note, look at page 1 to notice that $h_2[n]$ is causal, but the other two filters are not. However, stability in discrete time is not very restrictive because we may have access to "future" samples anyway. On

Reminder:

$$z_1 \cdot z_2 = r_1 e^{j\theta_1} \cdot r_2 e^{j\theta_2} = r_1 \cdot r_2 e^{j(\theta_1+\theta_2)}$$

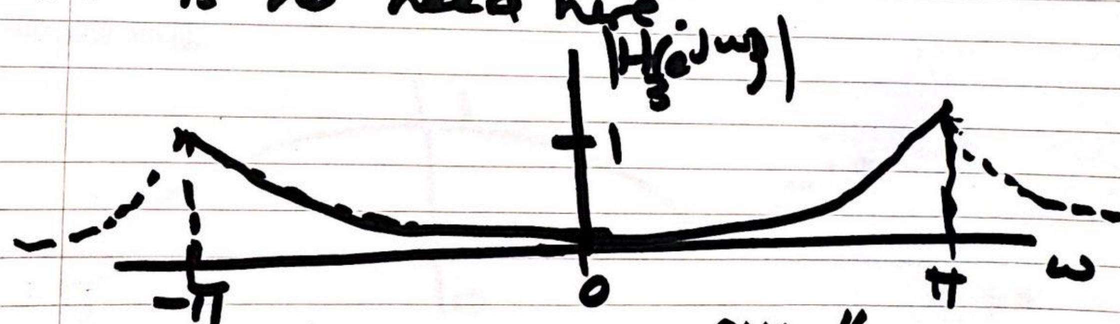
⑦

c-d

(ii) $H_3(e^{j\omega}) = \frac{1}{2} - \frac{1}{2}\cos\omega \quad -\pi \leq \omega < \pi$

$$|H_3(e^{j\omega})| = \left|\frac{1}{2} - \frac{1}{2}\cos\omega\right|$$

Note that $H_3(e^{j\omega}) \geq 0$ for this particular example so $|H_3(e^{j\omega})| = H(e^{j\omega})$. Normally, we would have to compute the $|\cdot|$ but there is no need here



"high-pass filter"

$\angle H_3(e^{j\omega}) = 0$ because it is purely real and positive.

$\angle H_3(e^{j\omega})$